

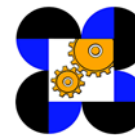


External Services

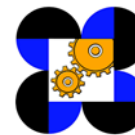
1. Measuring Instrument Calibration and Measurement Service

Metrological traceability to the International System of Units (SI) is provided to measuring instruments and samples used or manufactured in the country through calibration and measurement services.

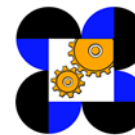
Office or Division:	National Metrology Division (NMD) – Industrial Technology Development Institute	
Classification:	Highly Technical	
Type of Transaction:	G2C - Government to Citizen, G2B - Government to Business, G2G - Government to Government	
Who may avail:	All	
CHECKLIST OF REQUIREMENTS		WHERE TO SECURE
F7.1.1-a Version 1 Technical Service Request (TSR) Form (one original, two photocopies)		Industrial Technology Development Institute / National Metrology Division / Receiving and Releasing Unit (RRU) All forms can also be downloaded from this link: https://bit.ly/40BskOK The online version of the Client Satisfaction Measurement (CSM) can be accessed using this link: https://tinyurl.com/NMDCSM
F7.1.1-b Version 2 On-site Technical Services Agreement Form (one original, one photocopy)		
F7.4.1 Version 1 Control Pass for Customer's Property Form (one original, one photocopy)		



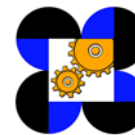
F8.6.2 Version 6 Client Satisfaction Measurement (CSM) (one original)	
F7.9 Version 1.0 NMD Complaint Form (one original)	
Measuring Instrument or Sample to be calibrated/tested	Customer
Measuring Instrument or Sample Accessories such as power cables, batteries, adapters, chargers, connectors, indicators, jigs, fittings, etc.	
Measuring Instrument or Sample Instructions/Operations Manual	
Measuring Instrument or Sample Special Requirements: • general – all battery-operated equipment and all main powered equipment ○ new/fresh batteries ○ power supply/charger/adapter ○ power cord/cable • road tanker ○ LTO Official Receipt (OR) ○ Certificate of Registration (CR) ○ clean internal walls of tanker • volumetric glassware ○ no deep scratches ○ no cracks	



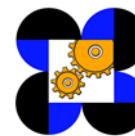
- clean and odorless
 - labeled for distinction (if submitted in bulk)
- **piston-operated pipette**
 - provided with original tips approved by the manufacturer
 - no residual liquids and dirt particles
 - no leaks
 - supplied with new batteries for digital pipettes
 - no cracks
 - not deformed, scratched, or heavily contaminated pipette shaft
- **test measure (calibration bucket)**
 - clean and odorless
 - no leaks
 - no internal and external damage, e.g., broken sight glass, rusty internal wall, deformed or dented
- **multi-delivery dispenser**
 - no residual liquids and dirt particles
 - no leaks
 - provided with original tips approved by the manufacturer
 - supplied with new batteries for motorized dispenser
 - no internal or external damage e.g., cracks, deformed, scratched, or heavily contaminated plug-in device for dispenser tip
- **hydrometer**
 - no deep scratches
 - no cracks
 - clean and odorless



- **proving tank**
 - clean and odorless
 - no leaks
 - no internal or external damage e.g., broken sight glass, rusty internal wall, severely deformed or dented
- **flowmeter**
 - includes a power supply for devices needing over 24 V
 - comes with new batteries for battery-operated devices
 - includes an operations manual
 - comes with fittings for calibration facility installation
- **pressure calibrator/transducer/ transmitter**
 - with power supply
 - supplied with new batteries if battery-operated
 - with clear indicator
 - standard connecting port (preferably NPT connections)
- **pressure balance / dead weight pressure balance**
 - provided with clean weights
 - sufficient, clean, and compatible fluid for cross-floating (for hydraulic type)
- **pressure gauge**
 - clean, clear glass and readable indications
 - standard connecting port (preferably NPT connections)
- **sphygmomanometer**



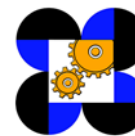
○ no visible cracks and lose-thread on the valve, cuff, and pneumatic connections ○ provided with cuff, inflation bulb, and standard valve ○ supplied with power supply/ charger or new batteries if automated ● load cell ○ with indicator, cables, and connectors ● force testing machines, push/pull gauge, dynamometer, crane scale ○ provided with jigs and connectors compatible with load cells ● gauge block ○ no rust ○ no prominent scratches on the measurements faces ● test weights ○ clean ○ free from dust, rust, and other contaminants ○ not corroded ○ must have a proper case with serial or control number ○ must bear permanent serial/control number for cast iron weights ○ must only have one hole used only as provision for adjustment for cast iron weights ● non-automatic weighing instruments ○ must bear a permanent serial or control number ● liquid-in-glass thermometers ○ no gaps in the liquid column	
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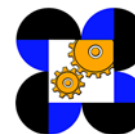
○ clear graduation/marking ○ clean ○ no bubbles ○ no cracks ● thermo-hygrometer ○ has an option to disable the auto-off function ○ provided with new batteries for digital thermo-hygrometer ● digital thermometer ○ clean ○ probe/s not defective or bent ○ clear display ○ provided with new batteries ● digital thermometer with thermocouple probe ○ thermocouple wire not bent ○ thermocouple wire at least 500 mm length ○ no head assembly ○ sensor diameter, 6 mm and below ● industrial platinum resistance thermometer ○ 4-wire ○ spade connectors ● infrared thermometer ○ with known emissivity ○ with a known spectral response range ○ with a known distance-to-space ratio ○ with center point measurement capability (thermal imager) ● wood moisture meter ○ provided with wood samples ● rice moisture meter	
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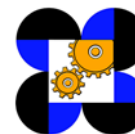
o supplied with five pieces (500 g per piece) of different moisture contents; vacuum-packed		Industrial Technology Development Institute / National Metrology Division /Cashier Section		
Authorization Letter addressed to the NMD Chief (one original)				
Invoice (one original)				
CUSTOMER STEPS	AGENCY ACTION	FEES TO BE PAID	PROCESSING TIME	PERSON RESPONSIBLE
1. Sign the Visitors' Logbook at the Security Guard's Lobby Desk.	1.1 Guard-on-Duty verifies that the customer has signed the Visitors' Logbook.	none	1 minute	Guard-on-Duty, Building Lobby, Lobby Desk, Metrology Building
2. Proceed to the Receiving Window of the RRU for assistance.	2.1 RRU Staff provides the customer with the forms for filling out and sends them for the NMD Technical Staff.	none	1 minute	Supervising Science Research Specialist, Receiving and Releasing Unit, Metrology Building
3. Fill out the forms. Using one of the computers in the RRU, access the National Metrology Laboratory Information Management System (NLIMS) and fill out the online forms completely as well.	3.1 The NMD Technical Staff inspects the suitability of the equipment/sample for calibration or testing and validates the forms.	none	13 minutes (This is the average time spent to inspect and validate one measuring instrument/ sample.)	Supervising Science Research Specialist, Receiving and Releasing Unit, Metrology Building
Request assistance in using	3.2 The NMD Technical			



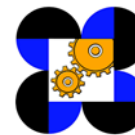
NLIMS if necessary. The following forms are required: For in-house services: TSR Form, Control Pass for Customer's Property Form. For on-site services: TSR Form, On-site Technical Services Agreement Form.	Staff prints the validated forms.			
4. Signify agreement to the Terms and Conditions of the Service and bring the validated forms to the Receiving Window of the RRU.	4.1 RRU Staff assigns a TSR number to the equipment/ sample and returns the validated forms to the customer.	none	1 minute	<i>Supervising Science Research Specialist, Receiving and Releasing Unit, Metrology Building</i>
5. Proceed to the ITDI Cashier for full payment of fees. If the customer requires a Statement of Account (SOA) prior to payment, proceed to the Accounting Section of the Financial Management Division to secure it. Refer to "Processing of Statement of Account."	5.1 ITDI Cashier processes payment of the customer Refer to "Issuance of Invoice (Over-the-Counter Payment)" or to "Issuance of Invoice (Direct Payment)."	Refer to 'NMD Schedule of Fees and Charges for Measuring Instrument Calibration and Measurement Services as of March 2025' after the service specification table.	5 minutes	<i>Administrative Officer III, Cashier Section, Administrative Division / 2nd Floor Metrology Building</i>



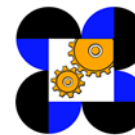
6. Present the Invoice and submit the NMD's copy of the TSR Form to the RRU Staff.	6.1 RRU Staff retrieves the NMD's copy of the TSR Form, then provides the customer with a copy of their form along with a receipt.	none	1 minute	<i>Supervising Science Research Specialist, Receiving and Releasing Unit, Metrology Building</i>
7. Surrender a copy of the Control Pass for Customer's Property Form to the Guard-on-Duty.	7.1 The Guard-on-Duty ensures the customer signs out properly in the Visitors' Logbook.	none	1 minute	<i>Guard-on-Duty, Building Lobby, Lobby Desk, Metrology Building</i>
8. Depart the premises of ITDI.	8.1 NMD Technical Staff receives the TSR Form and conducts the requested calibration and measurement service. For on-site services, provision of actual calibration and measurement service starts upon the arrival of the NMD Technical Staff at the location on the agreed date and time.	none	20 days (paused clock) (The processing time declared is the average time it takes to calibrate and/or test one measuring instrument).	<i>Supervising Science Research Specialist, Receiving and Releasing Unit, Metrology Building</i>



9. Sign the Visitors' Logbook at the Security Guard's Lobby Desk.	9.1 Guard-on-Duty verifies that the customer has signed the Visitors' Logbook.	none	1 minute	<i>Guard-on-Duty</i> , Building Lobby, Lobby Desk, Metrology Building
10. Proceed to the Releasing Window of the RRU for assistance and present the documents. For in-house services: Customer's copies of the TSR Form and Control Pass for Customer's Property Form. For on-site services: Customer's copy of the TSR Form.	10.1 RRU Staff reviews the documents to verify the status of the requested service.	none	2 minutes	<i>Supervising Science Research Specialist</i> , Receiving and Releasing Unit, Metrology Building
	10.2 For in-house services: RRU Staff returns the Control Pass for Customer's Property Form (customer's and NMD's copies) and equipment/sample. For onsite services: RRU Staff provides the receiving copy of the calibration certificate/test report and the CSM. (Customer executes Step 12)	none	1 minute	<i>Supervising Science Research Specialist</i> , Receiving and Releasing Unit, Metrology Building
11. Inspect the equipment/sample. If no irregularity is found, fill out the	11.1 RRU Staff provides the receiving copy of the	none	1 minute	<i>Supervising Science Research Specialist</i> , Receiving and Releasing



'Exit Pass' portion of the Control Pass for Customer's Property Forms, indicate in the 'Remark' the statement "Item(s) received in good condition.", and affix signature.	calibration certificate/test report and the CSM.			Unit, Metrology Building
12. Fill out the CSM and drop it in the Feedback Dropbox (or answer the online version of the CSM). Inspect the receiving copy of the calibration certificate/test report for any discrepancies. If no discrepancy is found, affix a signature on each page of the receiving copy and return it to the RRU Staff. Request for the NMD Complaint Form from the RRU staff or download it online for complaints.	12.1 RRU Staff hands over the original copy of the calibration certificate/test report to the customer.	none	1 minute	<i>Supervising Science Research Specialist,</i> Receiving and Releasing Unit, Metrology Building
13. Surrender the Control Pass for Customer's Property Forms, if applicable, to the Guard-on-Duty.	13.1 The Guard-on-Duty signs the Control Pass for Customer's Property Form, then provides the customer with their copy of the form.	none	1 minute	<i>Guard-on-Duty,</i> Building Lobby, Lobby Desk, Metrology Building



14. Depart the premises of ITDI.	14.1 RRU Staff encodes and archives the calibration certificate/test report.	none	1 minute	<i>Supervising Science Research Specialist, Receiving and Releasing Unit, Metrology Building</i>
TOTAL:		Refer to 'NMD Schedule of Fees and Charges for Measuring Instrument Calibration and Measurement Services as of March 2025' after the service specification table.	20 days, 31minutes	

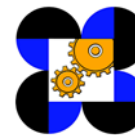
Measuring Instrument Calibration and Measurement Services qualified for multi-stage processing.



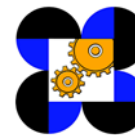
**NMD SCHEDULE OF FEES AND CHARGES FOR MEASURING INSTRUMENT CALIBRATION
AND MEASUREMENT SERVICES
As of March 2025**

Electricity, Time, Frequency, and Photometry

Type of Equipment / Device	Range or Capacity	Calibration Fee (PHP)
Conductivity Meter	0.1 μ S (minimum)	845.00
Centrifuge	Up to 18,000 r/min	800.00
DC Voltage Standard (per voltage level)	1.018 V and 10 V	5,590.00
Decade Resistance Box (per dial)	0 Ω to 100 M Ω	1,140.00
Double Bridge	Up to 100 M Ω	2,500.00
Earth Tester	Up to 100 M Ω	1,350.00
	per succeeding range	400.00
Frequency Calibrator	Up to 225 MHz	2,700.00
Frequency Counter	10 MHz	1,550.00
GPS Receiver	1 pps	9,000.00
Ground Strap Tester / Checker	Up to 100 M Ω	800.00
Illuminance / Lux / Light Meter	(380 to 2000) lux	2,600.00
Insulation Tester	Up to 1000 V	1,050.00
	per succeeding range	350.00
Kelvin Bridge	Up to 100 M Ω	2,500.00

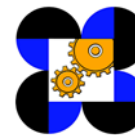


mA Calibrator	Up to 1000 mA	1,890.00
Megohmmeter / Megger Tester	Up to 1000 V	1,090.00
	per succeeding range	350.00
Multimeter, Digital	4 ½ digits; Up to 1000 Vdc; Up to 650 Vac; Up to 1 A AC/DC; Up to 100 MΩ	2,500.00
	6 ½ digits; Up to 1000 Vdc; Up to 650 Vac; Up to 1 A AC/DC; Up to 100 MΩ	17,000.00
pH Meter	0 to 14 pH	845.00
pH Simulator	0 to 14 pH	845.00
Process Calibrator (input/output)	Up to 1000 Vdc; Up to 650 Vac; Up to 1 A AC/DC; Up to 100 MΩ	4,900.00
Puncture Tester	Up to 5 kV AC/DC; Up to 1 A AC/DC; Up to 100 MΩ	1,700.00
Resistance Bridge / Wheatstone Bridge	1 Ω to 10 MΩ	1,700.00
Stopwatch / Timer	15 Minutes (minimum)	800.00
Standard Resistor (by Ratio)	1 Ω to 1 MΩ	6,500.00
Stroboscope	Up to 99,000 r/min	800.00
Tachometer (non-contact type)	Up to 18,000 r/min	800.00
Time Mark Generator	Up to 225 MHz	2,450.00



Pressure

Type of Equipment / Device	Range or Capacity	Calibration Fee (PHP)
Absolute Pressure Calibrator	0.1 MPa to 100 MPa	6,110.00
Absolute Pressure Industrial Gauge	0.1 MPa to 100 MPa	975.00
Absolute Pressure Test Gauge	0.1 MPa to 100 MPa	3,055.00
Hydraulic Pressure Balance / Deadweight Tester	0.1 MPa to 100 MPa	20,150.00
Hydraulic Pressure Calibrator	0.1 MPa to 100 MPa	6,110.00
Hydraulic Pressure Gauge	0.1 MPa to 100 MPa	975.00
Hydraulic Pressure Test Gauge	0.1 MPa to 100 MPa	3,055.00
Pneumatic Pressure Balance	0 bar to 40 bar	20,150.00
Deadweight Pressure Tester	0 bar to 40 bar	20,150.00
Pneumatic Pressure Calibrator	0 bar to 200 bar	6,110.00
Pneumatic Pressure Gauge	0 bar to 20 bar	975.00
Pneumatic Pressure Test Gauge	0 bar to 20 bar	3,055.00
Testing of Sphygmomanometer	0 mmHg to 250 mmHg	1,300.00
Vacuum Calibrator	0 bar to -1 bar	6,200.00
Vacuum Gauge	0 bar to -1 bar	975.00
Vacuum Test Gauge	0 bar to -1 bar	3,055.00

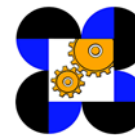


Force

Type of Equipment / Device	Range or Capacity	Calibration Fee (PHP)
Testing Machines	Up to 2000 kN (5 test pts. per range)	3,700.00
	per additional test point	750.00
Durometer A	0 HA to 90 HA	780.00
Durometer D	0 HD to 90 HD	1,560.00

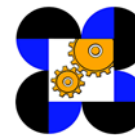
Length and Dimensional Metrology

Type of Equipment / Device	Range or Capacity	Calibration Fee per Piece (PHP)
Gauge Block Grade 0 (Steel)	0.5 mm to 100 mm	1,160.00
Gauge Block Grade 1 (Steel)		1,030.00
Gauge Block Grade 2 (Steel)		1,030.00



Contact Thermometry

Type of Equipment / Device	Range or Capacity	Calibration Fee (PHP)
Digital Thermometer	-70 °C to +250 °C (first 4 cal. points)	2,180.00
	per additional cal. point	650.00
	per additional probe	1,090.00
Industrial Platinum Resistance Thermometer	-70 °C to +250 °C (first 6 cal. points)	11,700.00
	per additional cal. point	975.00
Liquid-in-Glass Thermometer	-70 °C to +250 °C (first 4 cal. points)	2,180.00
	per additional cal. point	650.00
(Wall / Refrigerator / Bimetallic) Thermometer	-70 °C to +250 °C (first 4 cal. points)	1,820.00
	per additional cal. point	650.00
Thermocouple with Indicator	300 °C to 1000 °C (per cal. point)	2,150.00



Non-Contact Thermometry

Type of Equipment / Device	Range or Capacity	Calibration Fee (PHP)
Infrared Thermometer	Low-range: (20, 25, 30, 35, 50, 65, 70, 100) °C	2,800.00
	High-range: (50, 100, 150, 200, 250, 300, 350, 400) °C	2,800.00
	Body temperature-range: (30, 35, 37, 39, 42) °C	2,800.00

Enclosures

Type of Equipment / Device	Range or Capacity	Calibration Fee (PHP)
Furnace	300 °C to 1000 °C (first cal. point)	2,730.00
	per additional cal. point	1,400.00
Oven / Freezer / Incubator / Cold Storage / Walk-in Enclosures / Water Bath / Refrigerator, etc.	-70 °C to +250 °C (first cal. point)	2,730.00
	per additional cal. point	1,400.00



Humidity

Type of Equipment / Device	Range or Capacity	Calibration Fee (PHP)
Thermo-hygrometer	(20, 25, 30) °C; (40, 60 and 80) %rh at 23 °C	1,274.00
Thermo-hygrograph	(20, 25, 30) °C; (40, 60 and 80) %rh at 23 °C	3,822.00

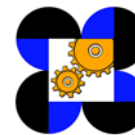
Moisture

Type of Equipment / Device	Range or Capacity	Calibration Fee (PHP)	Remarks
Wood Moisture Meter	9 % to 21 %	840.00	Test
Rice / Grain Moisture Meter	9 % to 21 %	2,500.00	Calibration; 5 samples of varying moisture content
Rice / Grain Moisture Meter	9 % to 21 %	1,800.00	Test; 1 sample



Volume

Type of Equipment / Device	Range or Capacity	Calibration Fee (PHP)
Burette	0.1 mL to 100 mL (5 cal. points)	689.00
Volumetric Flask	1 mL to 5000 mL	689.00
Multiple Delivery Dispenser	1 μ L to 100000 μ L (5 cal. points)	1,900.00
Piston Pipette	1 μ L to 10000 μ L (3 cal. points)	1,900.00
	per addition cal. point	630.00
Pipette	0.1 mL to 100 mL	689.00
Proving Tanks (Gravimetric)	up to 500 L	5,200.00
Proving Tanks (Volumetric)	100L to 400L	3,500.00
Proving Tanks (Volumetric)	>400L to 2000 L	3,500.00
Proving Tanks (Volumetric)	>2000 L to 5000 L	4,500.00
Road Tankers (Volume capacity determination)	up to 10000 L	920.00
	>10000 L to 15000 L	1,370.00
	>15000 L to 20000 L	1,840.00
	>20000 L to 25000 L	2,300.00
	>25000 L to 30000 L	2,750.00



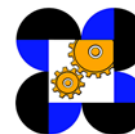
	>30000 L to 35000 L	3,200.00
	>35000 L to 40000 L	3,650.00
	>40000 L to 45000 L	4,100.00
	>45000 L to 50000 L	4,200.00
Test Measure (Gravimetric)	5 L , 10 L , 20 L	2,340.00
Test Measure (Volumetric)	5 L , 10 L	390.00
Test Measure (Volumetric)	20 L	780.00

Fluid Flow

Type of Equipment / Device	Range or Capacity	Calibration Fee (PHP)
Flowmeter	Water: 1 L/min to 2000 L/min (5 cal. points)	3,360.00
	Air: 0.2 m/s to 25 m/s (5 cal. points)	3,360.00

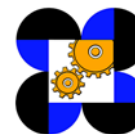
Density

Type of Equipment / Device	Range or Capacity	Calibration Fee (PHP)
Hydrometers	(600 to 2000) kg/m ³	1,200.00



Mass Standards

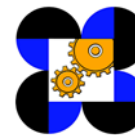
Class / Type	Range	Calibration Fee per Piece (PHP)	Remarks
OIML Class E2	1 mg to 50 g	1,090.00	DAkkS Certificate (1 mg to 1 kg only) NML Certificate (2 kg to 50 kg)
	100 g to 500 g	1,480.00	
	1 kg to 10 kg	1,660.00	
	20 kg to 50 kg	1,990.00	
OIML Class F1 / F2	1 mg to 50 g	600.00	DAkkS Certificate (1 mg to 50 kg only) NML Certificate (100 kg to 200 kg)
	100 g to 500 g	650.00	
	1 kg to 10 kg	950.00	
	20 kg to 50 kg	1,050.00	
	100 kg to 200 kg	2,100.00	
OIML Class M1 / M2 / M3	1 mg to 500 g	450.00	DAkkS Certificate (1 mg to 50 kg only) NML Certificate (100 kg to 200 kg)
	1 kg to 10 kg	600.00	
	20 kg to 50 kg	700.00	
	100 kg to 200 kg	1,500.00	
	500 kg	2,100.00	
Free Nominal (Stainless Steel)	1 mg to 50 g	600.00	DAkkS Certificate



	> 50 g to 500 g	650.00	(1 mg to 50 kg only)
	> 500 g to 10 kg	950.00	NML Certificate (100 kg to 200 kg)
	> 10 kg to 50 kg	1,050.00	
	> 50 kg to 200 kg	2,100.00	
Free Nominal (Other Materials)	1 mg to 500 g	450.00	DAkkS Certificate (1 mg to 50 kg only)
	> 500 g to 10 kg	600.00	
	> 10 kg to 50 kg	700.00	
	> 50 kg to 200 kg	1,500.00	NML Certificate (100 kg to 500 kg)
	> 200 kg to 500 kg	2,100.00	

Nonautomatic Weighing Instruments (NAWI)

Type of Service	Range / Accuracy Class	Calibration Fee per Piece (PHP)	Remarks
Calibration of NAWI, electronic type only (on-site calibration)	Up to 2 kg (using OIML Class E ₂)	2,000.00	DAkkS Certificate (up to 600 g only)
			NMD Certificate (>600 g up to 2 kg)
	Up to 60 kg (using OIML Class F ₁)	1,530.00	DAkkS Certificate (up to 20 kg only)



			NMD Certificate (>20 kg up to 60 kg)
	Up to 200 kg (using OIML Class F ₂)	1,530.00	DAkkS Certificate (up to 60 kg only)
			NMD Certificate (>60 kg up to 200 kg)
	Up to 300 kg (using OIML Class M ₁)	1,530.00	NMD Certificate
Testing of NAWI, electronic and mechanical types	High Accuracy* (using OIML Class F ₂ ; 1 tonne)	1,200.00	NMD Certificate
	Medium Accuracy* & Ordinary Accuracy* (using OIML Class M ₁ and substitution material; 1 tonne)	1,080.00	
	*per additional tonne	20.00	

On-Site Calibration Service

Within 50 km radius from NML per day per team	PHP 2,000.00
More than 50 km radius from NML per day per team	PHP 3,000.00